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(54) TRAVEL PLAN GENERATION DEVICE AND TRAVEL PLAN GENERATION METHOD

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(57) **ABSTRACT**

A travel plan generation device includes a controller configured to: generate a travel plan in which a vehicle passes through a first supply facility for supplying energy of the vehicle; acquire first supply availability information indicating a vacancy state of a first supply facility after the vehicle starts traveling based on the generated travel plan; notify notification information indicating that there is no available charging station; determine whether or not supply of the energy is available at the first supply facility based on the first supply availability information of the first supply facility; and when determining that the supply of the energy is not available at the first supply facility, identify a second supply facility different from the first supply facility and generate the travel plan in which the vehicle passes through the second supply facility instead of the first supply facility.

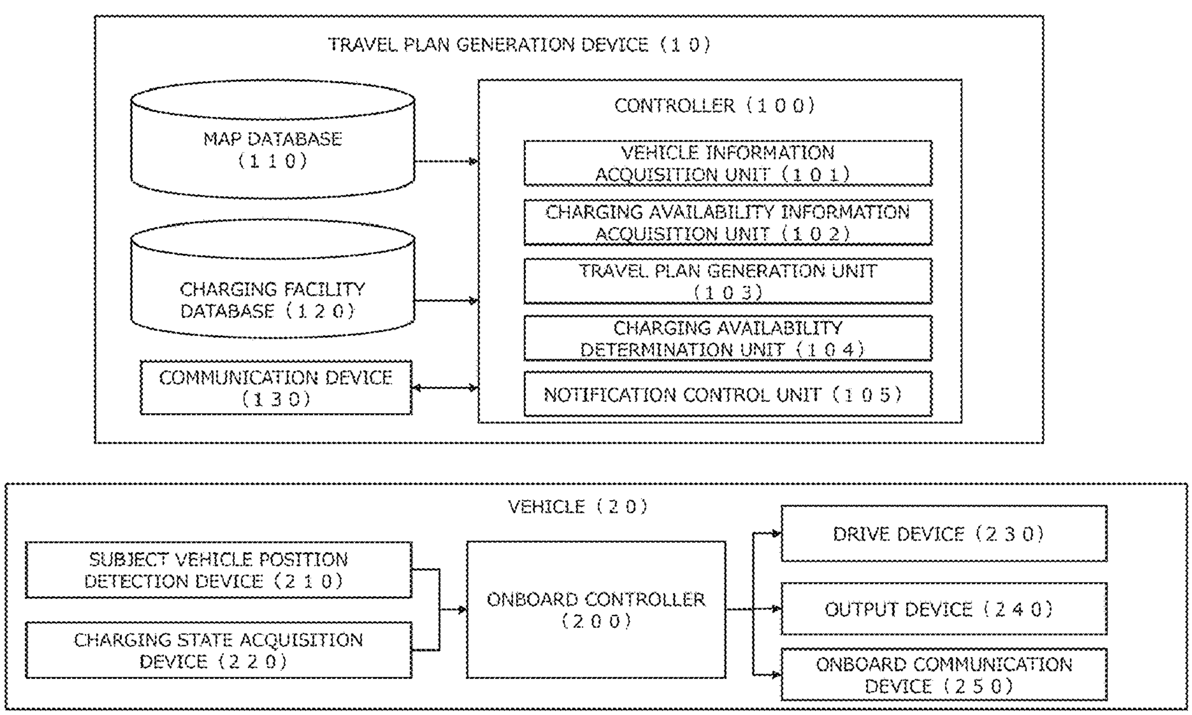
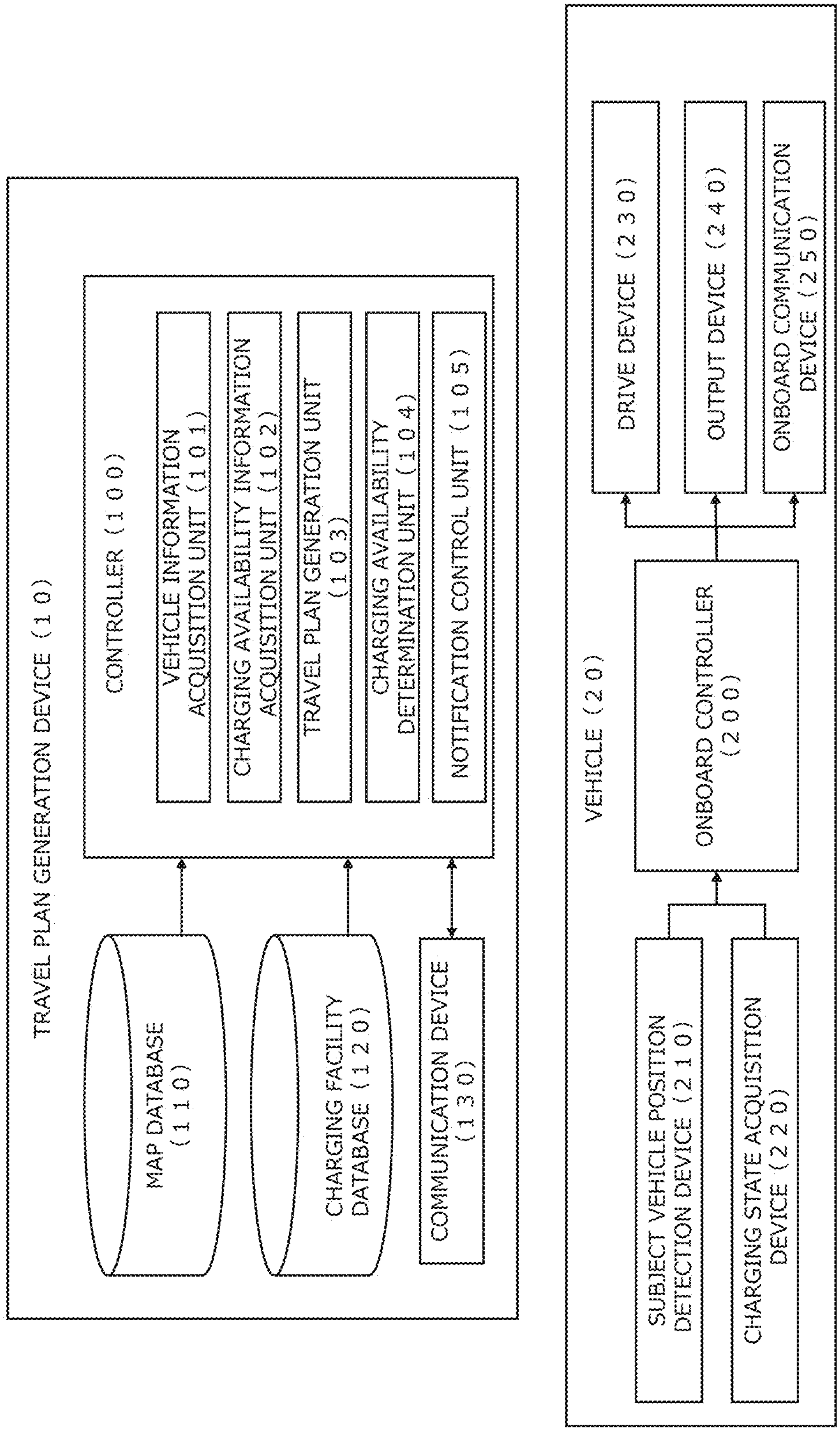


FIG. 1



TRAVEL PLAN GENERATION DEVICE (10)
CONTROLLER (100)
VEHICLE INFORMATION ACQUISITION UNIT (101)
CHARGING AVAILABILITY INFORMATION ACQUISITION UNIT (102)
TRAVEL PLAN GENERATION UNIT (103)
CHARGING AVAILABILITY DETERMINATION UNIT (104)
NOTIFICATION CONTROL UNIT (105)
MAP DATABASE (110)
CHARGING FACILITY DATABASE (120)
COMMUNICATION DEVICE (130)
VEHICLE (20)
ONBOARD CONTROLLER (200)
SUBJECT VEHICLE POSITION DETECTION DEVICE (210)
CHARGING STATE ACQUISITION DEVICE (220)
DRIVE DEVICE (230)
OUTPUT DEVICE (240)
ONBOARD COMMUNICATION DEVICE (250)

FIG. 2

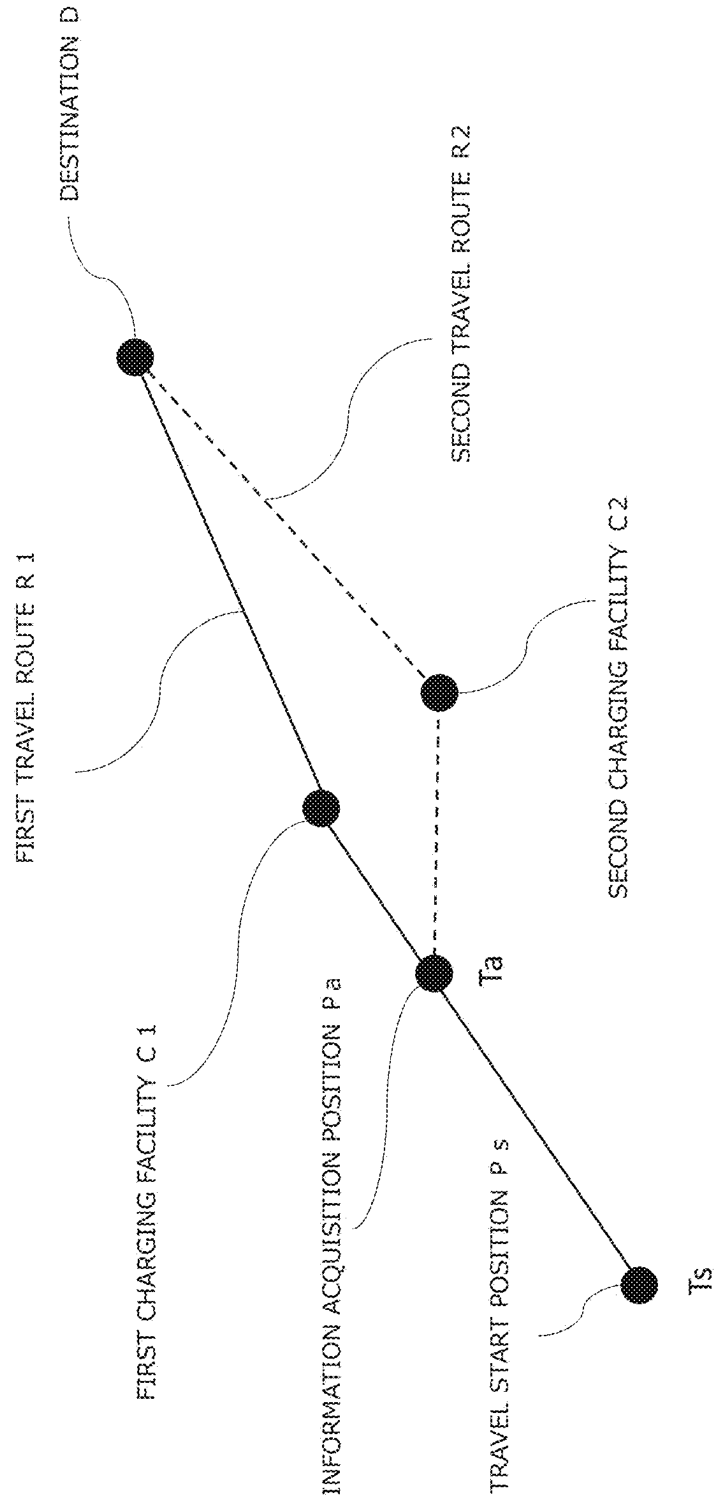
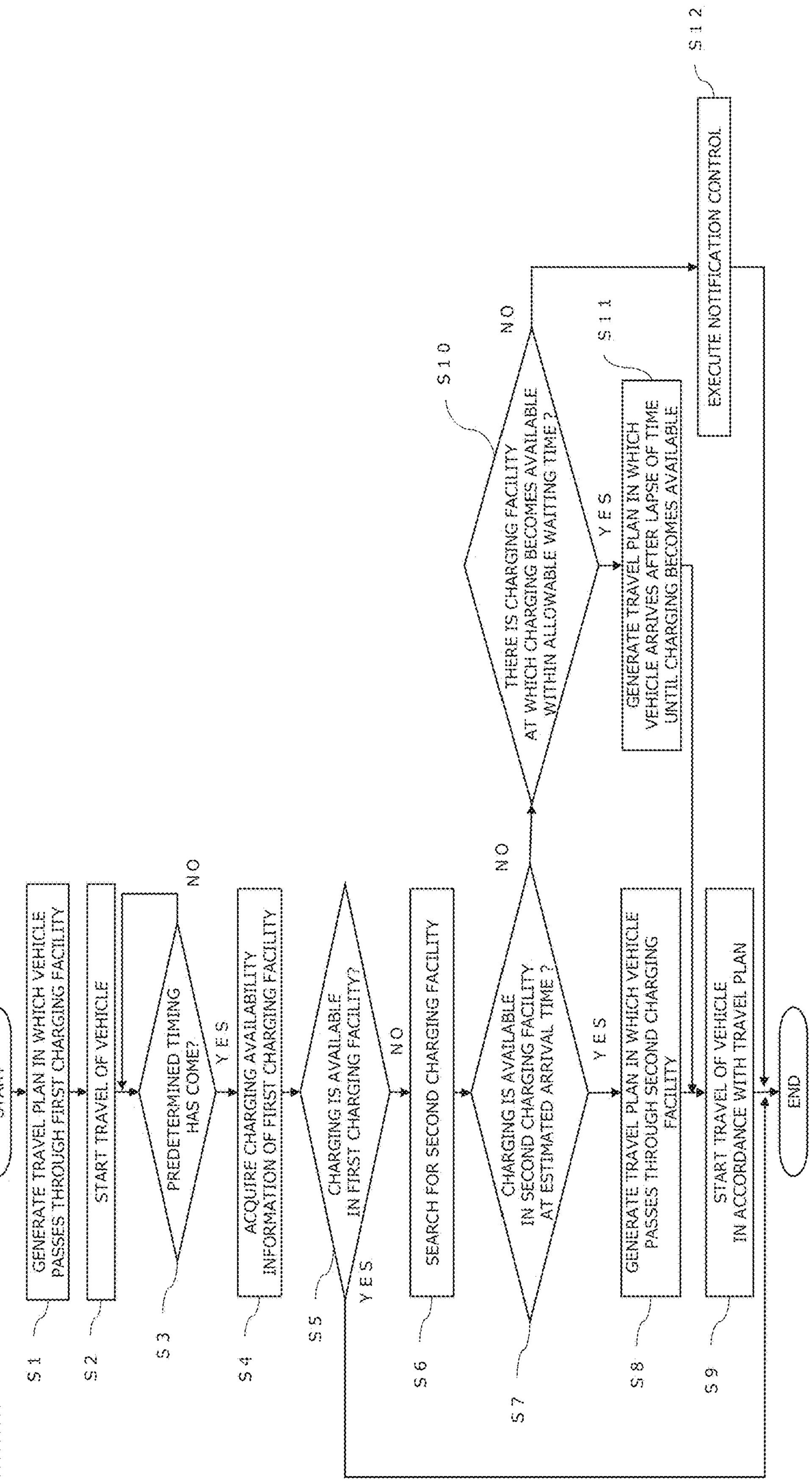

DESTINATION D
SECOND TRAVEL ROUTE R 2
FIRST TRAVEL ROUTE R 1
SECOND CHARGING FACILITY C 2
Ta
FIRST CHARGING FACILITY C 1
INFORMATION ACQUISITION POSITION P a
TRAVEL START POSITION P s
Ts

FIG. 3


START
S1
GENERATE TRAVEL PLAN IN WHICH VEHICLE PASSES THROUGH FIRST CHARGING FACILITY
S2
START TRAVEL OF VEHICLE
S3
PREDETERMINED TIMING HAS COME?
NO
YES
S4
ACQUIRE CHARGING AVAILABILITY INFORMATION OF FIRST CHARGING FACILITY
S5
CHARGING IS AVAILABLE IN FIRST CHARGING FACILITY?
YES
NO
S6
SEARCH FOR SECOND CHARGING FACILITY
S7
CHARGING IS AVAILABLE IN SECOND CHARGING FACILITY AT ESTIMATED ARRIVAL TIME ?
YES
NO
S8
GENERATE TRAVEL PLAN IN WHICH VEHICLE PASSES THROUGH SECOND CHARGING FACILITY
S9
START TRAVEL OF VEHICLE IN ACCORDANCE WITH TRAVEL PLAN
END
S10
THERE IS CHARGING FACILITY AT WHICH CHARGING BECOMES AVAILABLE WITHIN ALLOWABLE WAITING TIME ?
YES
NO
S11
GENERATE TRAVEL PLAN IN WHICH VEHICLE ARRIVES AFTER LAPSE OF TIME UNTIL CHARGING BECOMES AVAILABLE
S12
EXECUTE NOTIFICATION CONTROL

TRAVEL PLAN GENERATION DEVICE AND TRAVEL PLAN GENERATION METHOD

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] Priority is claimed on U.S. patent application Ser. No. 17/999,472 filed on Nov. 21, 2022, which is a National Stage Entry of International Patent Application No. PCT/JP2021/010909 filed on Mar. 17, 2021, claiming foreign priority from Japanese Patent Application No. 2020-089620 filed in Japan on May 22, 2020, the contents of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] The present invention relates to a travel plan generation device and a travel plan generation method.

[0003] The present application claims priority based on Japanese Patent Application No. 2020-089620 filed on May 22, 2020. For those designated countries which permit the incorporation by reference, the content described and/or illustrated in the above application is incorporated by reference in the present application as part of the description and/or drawings of the present application.

BACKGROUND ART

[0004] A technique is known, which determines a recommended charging pattern defined by at least one recommended charging facility on a travel route from a present position to a destination of an electric vehicle and a recommended charging amount in the recommended charging facility (see Patent Document 1: JP2013-11458, for example).

[0005] In the technique according to Patent Document 1, the recommended charging pattern is determined on the basis of the travel route from the present position to the destination of an electric vehicle, the remaining battery capacity, the position of a charging facility on the travel route, and the allowable remaining capacity range. The allowable remaining capacity range is a range of possible values for the remaining battery capacity and is set in consideration of an influence of the remaining battery capacity on the battery life.

[Prior Art Document] [Patent

Document] [Patent

[0006] Document 1] JP2013-11458

SUMMARY OF INVENTION

Problems to be Solved by Invention

[0007] However, in Patent Document 1, there is a problem that the technique cannot change the charging facility through which the vehicle passes even when supply of energy to the vehicle becomes non-available at the recommended charging facility after the vehicle starts traveling toward the destination until the vehicle arrives at the recommended charging facility.

[0008] A problem to be solved by the present invention is to provide a travel plan generation device and a travel plan generation method which can change the charging facility through which the vehicle passes when the supply of the energy to the vehicle becomes non-available at the charging facility after the vehicle starts traveling until the vehicle arrives at the charging facility through which the vehicle is scheduled to pass.

Means for Solving Problems

[0009] The present invention solves the above problem thorough generating a travel plan in which a vehicle passes through a first supply facility for supplying energy for the vehicle, acquiring supply availability information on energy of a first supply facility at a predetermined timing after the vehicle starts traveling based on the generated travel plan, determining whether or not supply of energy is available at the first supply facility based on the supply availability information, and identifying a second supply facility which is different from the first supply facility when determining that the supply of the energy is not available at the first supply facility, and generating a travel plan in which the vehicle passes through the second supply facility instead of the first supply facility.

Effect of Invention

[0010] According to the present invention, the charging facility through which the vehicle passes can be changed when the supply of the energy for the vehicle becomes non-available at the charging facility after the vehicle starts traveling until the vehicle arrives at the charging facility through which the vehicle is scheduled to pass.

BRIEF DESCRIPTION OF DRAWINGS

[0011] FIG. **1** is a block diagram illustrating an example of a travel plan generation device according to one or more embodiments of the present invention.

[0012] FIG. **2** is a diagram illustrating an example of generating of a travel plan according to one or more embodiments of the present invention.

[0013] FIG. **3** is a flowchart of a travel plan generation method according to one or more embodiments of the present invention.

MODE(S) FOR CARRYING OUT THE INVENTION

[0014] Hereinafter, one or more embodiments of a travel plan generation device according to the present invention will be described with reference to the drawings. The travel plan generation device according to one or more embodiments is a device for generating a travel plan in which a vehicle passes through a supply facility for supplying energy for the vehicle.

[0015] The configuration of the travel plan generation device according to one or more embodiments will be described with reference to FIG. **1**. FIG. **1** is a block diagram illustrating an example of a travel plan generation device **10** according to one or more embodiments. As illustrated in FIG. **1**, the travel plan generation device **10** according to one or more embodiments is connected to a vehicle **20** via a network constituting a telecommunication network. The travel plan generation device **10** may be connected to a plurality of vehicles. The travel plan generation device **10** is connected to an energy supply facility via a network. The travel plan generation device **10** may be connected to a plurality of energy supply facilities. The travel plan generation device **10** generates a travel plan in which the vehicle

20 travels from the present position to the destination through the energy supply facility based on vehicle information of the vehicle **20**, information of the destination, and information of the energy supply facility. As illustrated in FIG. **1**, the travel plan generation device **10** includes at least a controller **100**, a map database **110**, a charging facility database **120**, and a communication device **130**. Note that, in one or more embodiments of the present invention, an electric vehicle using, as a drive source, a motor driven by electric power supplied from a battery is described as an example of the vehicle **20**, but it may be an automobile other than an electric vehicle. For example, the vehicle **20** may be an automobile using a gasoline engine as a drive source, or a hybrid automobile using both an engine and a motor as a drive source. When the vehicle **20** is an electric vehicle, the energy is described as an electric power and the energy supply facility is described as a charging facility. If the vehicle **20** is an automobile having an engine as a drive source, the energy may be used as a gasoline fuel, and the energy supply facility may be used as an oil supply facility.

[0016] The controller **100** includes a computer having hardware and software, the computer including a ROM (Read Only Memory) storing a program, a CPU (Central Processing Unit) executing the program stored in the ROM, and a RAM (Random Access Memory) functioning as an accessible storage device. As the operation circuitry, a MPU (Micro Processing Unit), DSP (Digital Signal Processor), ASIC (Application Specific Integrated Circuit), FPGA (Field Programmable Gate Array), or the like can be used instead of or together with CPU. The controller **100** includes, as functional blocks, a vehicle information acquisition unit **101**, a charging availability information acquisition unit **102**, a travel plan generation unit **103**, a charging availability determination unit **104**, and a notification control unit **105**. The controller **100** executes each function in cooperation with software for implementing each of the above-described functions or executing each process, and hardware. Specifically, the controller **100** first acquires vehicle information of the vehicle **20**, destination information, and charging facility information. Then, the controller **100** identifies the charging facility for charging the vehicle **20** as a first charging facility based on the vehicle information of the vehicle **20**, the destination information, and the charging facility information. The controller **100** generates a travel plan in which the vehicle **20** travels from the present position to the destination through the first charging facility. When the charging of the vehicle **20** at the first charging facility becomes non-available after the vehicle **20** starts traveling in accordance with the generated travel plan until the vehicle **20** arrives at the first charging facility, the controller **100** identifies a charging facility different from the first charging facility as the second charging facility. Then, the controller **100** generates a travel plan in which the vehicle **20** passes through the second charging facility instead of the first charging facility. The first charging facility is a charging facility identified before or at the start of traveling of the vehicle **20**. The second charging facility is a charging facility identified as a replacement facility of the first charging facility after the vehicle **20** starts traveling. Note that, in one or more embodiments of the present invention, the functions of the controller **100** are divided into five blocks, and the functions of the respective functional blocks are described, but the functions of the controller **100** are not necessarily divided into five blocks. The functions of the controller **100** may be divided into four or less functional blocks or six or more functional blocks. In one or more embodiments of the present invention, the controller **100** executes the respective functions as the configuration of the travel plan generation device **10**, but the present invention is not limited. The respective functions of the controller **100** may be executed by the vehicle **20**. For example, the travel plan generation device **10** includes a charging facility database **120** and manages updating of the charging facility database **120** and so on. Then, the onboard controller **200** of the vehicle **20** may acquire charging availability information from the charging facility database **120**, determine whether or not charging is available at the charging facility, and generate a travel plan in which the vehicle **20** passes through the charging facility.

[0017] The vehicle information acquisition unit **101** acquires vehicle information of the vehicle **20** from the vehicle **20**. The vehicle information includes at least position information of the vehicle **20** and battery information of the vehicle **20**. The position information of the vehicle **20** is information indicating the present position of the vehicle **20**. The battery information is information indicating the remaining capacity of the battery of the vehicle **20**. In one or more embodiments of the present invention, the vehicle information acquisition unit **101** acquires the vehicle information of the vehicle **20** before the start of traveling of the vehicle **20** and during the travel of the vehicle **20**. The vehicle information of the vehicle **20** is transmitted to the travel plan generation device **10** via an onboard communication device **250**.

[0018] The charging availability information acquisition unit **102** acquires charging availability information of the charging facility. Specifically, the charging availability information acquisition unit **102** acquires from the charging facility database **120** full/vacant information indicating a vacancy state of a charging station in the charging facility as the charging availability information. The charging facility may have two or more charging stations. In one or more embodiments of the present invention, the charging availability information acquisition unit **102** acquires charging availability information of a first charging facility at a predetermined timing after the vehicle **20** starts traveling based on the travel plan. First, the charging availability information acquisition unit **102** sets, as the predetermined timing, a time point at which the vehicle **20** has reached a position within a predetermined range from the first charging facility. Then, the charging availability information acquisition unit **102** acquires the charging availability information of the first charging facility at the predetermined timing, that is, at a time point when the vehicle **20** has reached a position within a predetermined range from the first charging facility. The position within the predetermined range from the first charging facility is, for example, a point which is a predetermined distance away from the first charging facility or a point from which it takes a predetermined travel time to the first charging facility. The predetermined distance is, for example, distances of several kilometers to several tens of kilometers from the first charging facility. The predetermined travel time is, for example, several minutes to several tens of minutes. Based on the position information of the vehicle **20**, the map information, and the position information of the first charging facility, the charging availability information acquisition unit **102** determines whether or not the vehicle **20** has reached a point which is a predetermined distance away from the first charging facility. The charging availability information acquisition unit **102** may determine whether or not the vehicle **20** has reached a point from which it takes a predetermined travel time to the first charging facility based on the distance to the first charging facility and the vehicle speed of the vehicle **20**. The charging availability information acquisition unit **102** may set an interval between the predetermined timing to a predetermined interval. In this case, the charging availability information acquisition unit **102** acquires the charging availability information of the first charging facility at the predetermined intervals.

[0019] The charging availability information acquisition unit **102** may acquire, as the charging availability information of the first charging facility, full/vacant information indicating a vacancy state of a charging station in the first charging facility at a predetermined timing. The charging availability information acquisition unit **102** may acquire, as the charging availability information of the first charging facility, the full/vacant information indicating the vacancy state of the charging station in the first charging facility at a scheduled arrival time at which the vehicle **20** arrives at the first charging facility. Specifically, at first, the charging availability information acquisition unit **102** calculates the distance between the vehicle **20** and the first charging facility based on the position information of the vehicle **20** and the position information of the first charging facility, and calculates the scheduled arrival time of the vehicle **20** based on the calculated distance and the vehicle speed of the vehicle **20**. Then, the charging availability information acquisition unit **102** acquires, as the charging availability information of the first charging facility, the full/vacant information indicating the vacancy state of the charging station in the first charging facility at the calculated scheduled arrival time. At this time, the charging availability information acquisition unit **102** acquires, as the full/vacant information of the first charging facility at the calculated scheduled arrival time, information on whether or not there are reservations for use of the charging station in the first charging facility at the scheduled arrival time and whether or not the charging of the vehicle being charged at the first charging facility is completed by the scheduled arrival time. The reason why the information on the reservation for use of the scheduled arrival time is acquired is that the vehicle **20** cannot be charged if the charging station is reserved for use when the vehicle **20** arrives, even if the charging station is in the vacancy state at the time of the predetermined timing. The reason why the information on whether or not the charging of the vehicle being charged is completed by the scheduled arrival time is acquired is that even if the charging station is in the full state at the time of the predetermined timing, the vehicle **20** can be charged if the charging station is not in use for charging at the time when the vehicle **20** arrives. Then, if the charging is completed in one of the vehicles at the first charging facility by the estimated arrival time of the vehicle **20**, the charging availability information acquisition unit **102** acquires the full/vacant information indicating that there is a vacant space in the charging station of the first charging facility at the estimated arrival time.

[0020] Further, the charging availability information acquisition unit **102** acquires charging availability information of the second charging facility at the estimated arrival time at which the vehicle **20** arrives at the second charging facility. That is, the charging availability information acquisition unit **102** acquires full/vacant information indicating a vacancy state of the charging station in the second charging facility at the estimated arrival time. The charging availability information acquisition unit **102** acquires the charging availability information of the second charging facility after the estimated arrival time. The charging availability information of the second charging facility after the estimated arrival time is the full/vacant information indicating the vacancy state of the charging station in the second charging facility after the estimated arrival time. Specifically, the charging availability information acquisition unit **102** acquires, as the full/vacant information of the second charging facility after the estimated arrival time, information on whether or not there is a reservation for use of the charging station in the second charging facility after the estimated arrival time, and information on the time period until the charging of the vehicle being charged in the second charging facility is completed after the estimated arrival time. The charging availability information acquisition unit **102** may acquire the full/vacant information of the first charging facility after the time when the vehicle **20** arrives at the first charging facility.

[0021] The travel plan generation unit **103** generates a travel plan in which the vehicle **20** travels from the present position to the destination through the charging facility. First, the travel plan generation unit **103** calculates a travel route from the present position of the vehicle **20** to the destination based on the position information and the destination information of the vehicle **20**. The destination information is information on the destination of the occupant of the vehicle **20** and is set by the occupant. The travel route calculated at this time may be a travel route in which the travel time from the present position to the destination is shortest, or may be a travel route in which the travel distance from the present position to the destination is shortest. Next, the travel plan generation unit **103** identifies a charging facility located on the calculated travel route as the first charging facility. Specifically, the travel plan generation unit **103** identifies a range of a travelable distance with the remaining battery capacity of the vehicle **20** based on the battery information of the vehicle **20**. Then, the travel plan generation unit **103** identifies, as the first charging facility, a charging facility located on the travel route within the range based on the map information and the position information of the charging facilities. In addition, when there are a plurality of charging facilities located on the travel route within the range, the travel plan generation unit **103** identifies, as the first charging facility, a charging facility having the shortest travel distance or travel time from among the plurality of charging facilities. The travel plan generation unit **103** may identify not only a charging facility located on the travel route but also a charging facility located in a range of a predetermined distance from the travel route as the first charging facility. In this case, the travel plan generation unit **103** changes the travel route to a travel route in which the vehicle **20** passes through the identified first charging facility. As described above, the travel plan generation unit **103** generates the travel plan in which the vehicle **20** travels on the travel route from the present position to the destination.

[0022] The travel plan generation unit **103** may identify, as the first charging facility, a charging facility at which the charging of the vehicle **20** is available based on the charging availability information of the charging facility stored in the charging facility database **120**. For example, the travel plan generation unit **103** acquires the charging availability information of the charging facility located within the range of the travelable distance with the remaining battery capacity of the vehicle **20**, and identifies a charging facility at which the charging of the vehicle **20** is available as the first charging facility based on the charging availability information.

[0023] After the vehicle **20** starts traveling based on the generated travel plan, when the charging availability determination unit **104** determines that the charging of the vehicle **20** is not available at the first charging facility, the travel plan generation unit **103** generates a travel plan in which the vehicle **20** passes through the second charging facility instead of the first charging facility. Specifically, the travel plan generation unit **103** first searches for a second charging facility located within a predetermined distance from the first charging facility based on the position information of the first charging facility, the map information of the map database **110**, and the charging facility information of the charging facility database **120**. Then, the travel plan generation unit **103** identifies the second charging facility located within a predetermined distance from the first charging facility. At this time, the travel plan generation unit **103** may identify the second charging facility in which the charging of the vehicle **20** is available based on the charging availability information of the charging facility acquired by the charging availability information acquisition unit **102**. The acquired charging availability information of the second charging facility is charging availability information at an estimated arrival time at which the vehicle **20** arrives at the second charging facility. Further, the travel plan generation unit **103** may identify the second charging facility according to preference of the occupant. For example, the preference of the occupant may be set by the occupant in advance, or may be estimated by a past history. For example, the travel plan generation unit **103** identifies, as the preference of the preference, a charging facility having a wide road width up to the charging facility as the second charging facility. Then, the travel plan generation unit **103** calculates a travel route on which the vehicle **20** travels from the present position to the destination through the second charging facility based on the position information of the vehicle **20**, the destination information, and the position information of the second charging facility. The travel route calculated at this time may be a travel route in which the travel time from the present position to the destination is the shortest, or may be a travel route in which the travel distance from the present position to the destination is the shortest. As described above, the travel plan generation unit **103** generates the travel plan in which the vehicle **20** travels on the travel route from the present position to the destination.

[0024] Further, when the following determination is made by the charging availability determination unit **104**, the travel plan generation unit **103** generates a travel plan in which the vehicle **20** arrives at the second charging facility after a lapse of time period from the estimated arrival time at which the vehicle **20** arrives at the second charging facility until the charging the of the vehicle **20** becomes available. Specifically, the charging availability determination unit **104** determines that the charging of the vehicle **20** is available at the second charging facility within an allowable waiting time period from the estimated arrival time based on the charging availability information of the second charging facility after the estimated arrival time. The allowable waiting time period is a time period set by the occupant. For example, the occupant sets a waiting time period allowed by the occupant himself/herself as the allowable waiting time period. Similarly to the second charging facility, the travel plan generation unit **103** may generate a travel plan in which the vehicle **20** arrives at the first charging facility after a lapse of time period from the estimated arrival time until the charging of the vehicle **20** becomes available, when determining that the charging is available at the first charging facility within the allowable waiting time period from the estimated arrival time. Further, the travel plan generation unit **103** may identify, as the second charging facility, a charging facility at which the charging of the vehicle **20** is available within the allowable waiting time period from the estimated arrival time, based on the determination of the availability of each charging facility located within the predetermined range from the first charging facility.

[0025] Further, as the travel control for controlling the vehicle **20** to arrive at the first charging facility or the second charging facility after the lapse of time period from the estimated arrival time until the charging of the vehicle **20** becomes available, the travel plan generation unit **103** selects a standby place of the vehicle **20** from the standby available place, and generates a travel plan in which the vehicle **20** stops at the standby place for a period of time until the charging of the vehicle **20** becomes available. The standby available place is registered in the map database **110** in advance. When the time period until the charging of the vehicle **20** becomes available is 5 minutes, the travel plan generation unit **103** generates the travel plan in which the vehicle **20** stops at the standby place for 5 minutes. Further, the travel plan generation unit **103** may generate a detour route for the vehicle **20** to detour, and may generate the travel plan in which the vehicle **20** travels the detour route for a period of time until the charging of the vehicle **20** becomes available.

[0026] FIG. **2** is a diagram illustrating an example of generating of a travel plan according to one or more embodiments of the present invention. FIG. **2** shows the position of the vehicle **20** at each time and the travel plan of the vehicle **20**. In FIG. **2**, time Ts is a time at which the vehicle **20** starts traveling. The vehicle **20** is located at the travel start position Ps at the time Ts. At this time, the travel plan generation device **10** generates the first travel route R**1** based on the position information of the travel start position Ps, the destination D, and the first charging facility C**1**. Then, the vehicle **20** start traveling based on the first travel route R**1**. Then, at a predetermined timing Ta after starting the travel, the travel plan generation device **10** acquires the charging availability information of the first charging facility C**1**. When determining that the charging of the vehicle **20** is not available at the first charging facility C**1** based on the charging availability information of the first charging facility C**1**, the travel plan generation device **10** generates the second travel route R**2** in which the vehicle **20** travels from the information acquisition position Pa of the vehicle **20** to the destination D through the second charging facility C**2**. Note that, in FIG. **2**, an example in which one charging facility is present on the travel route is illustrated, but the present invention is not limited, and the vehicle **20** may pass through a plurality of charging facilities on the travel route. In this case, the travel plan generation device **10** acquires the charging availability information for each charging facility, and determines whether or not the charging of the vehicle **20** is available at the charging facility.

[0027] The charging availability determination unit **104** determines whether or not the charging of the vehicle **20** is available at the charging facility based on the charging availability information of the charging facility acquired by the charging availability information acquisition unit **102**. For example, when the charging availability information acquisition unit **102** acquires the full/vacant information indicating that there is no available space in the charging station of the charging facility, the charging availability determination unit **104** determines that the charging of the vehicle **20** is not available at the charging facility. In addition, when the charging availability information acquisition unit **102** acquires the full/vacant information indicating that there is a vacant space in the charging station of the charging facility, the charging availability determination unit **104** determines that the charging of the vehicle **20** is available at the charging facility. In one or more embodiments according to the present invention, the charging availability determination unit **104** determines whether or not the charging of the vehicle **20** is available at the first charging facility based on the charging availability information of the first charging facility acquired by the charging availability information acquisition unit **102**. Further, the charging availability determination unit **104** determines whether or not the charging of the vehicle **20** is available at the second charging facility at the estimated arrival time based on the charging availability information of the second charging facility at the estimated arrival time at which the vehicle **20** arrives at the second charging facility. Further, when determining that the charging of the vehicle **20** is not available at the second charging facility at the estimated arrival time, the charging availability determination unit **104** determines whether or not the charging of the vehicle **20** is available at the second charging facility within the allowable waiting time period from the estimated arrival time based on the charging availability information of the second charging facility after the estimated arrival time. Specifically, the charging availability determination unit **104** compares the time period until the charging of the vehicle **20** becomes available at the second charging facility with the allowable waiting time period. When the time period until the charging of the vehicle **20** becomes available is shorter than the allowable waiting time period, the charging availability determination unit **104** determines that the charging of the vehicle **20** at the second charging facility becomes available within the allowable waiting time period from the estimated arrival time. Further, the charging availability determination unit **104** determines whether or not the charging of the vehicle **20** becomes available at the first charging facility within the allowable waiting time period based on the charging availability information of the first charging facility after the time when the vehicle **20** arrives at the first charging facility. Further, the charging availability determination unit **104** may determine whether or not the charging of the vehicle **20** becomes available within the allowable waiting time period from the estimated arrival time for each charging facility located in a predetermined range from the first charging facility.

[0028] When determining that there is no charging facility at which the charging of the vehicle **20** is available within the allowable waiting time period from the estimated arrival time, the notification control unit **105** transmits to the vehicle **20** notification information indicating that there is no charging facility at which the charging of the vehicle **20** is available within the allowable waiting time period from the estimated arrival time. The notification information may include information on a waiting time period until the charging of the vehicle **20** becomes available at the second charging facility. Further, the notification information may include proposal information suggesting that the vehicle **20** stops at the standby place or travels on the detour route during the waiting time period.

[0029] The map database **110** is map information including road information, and stores the road information including, for example, an intersection or a branch point as a node and a road section between a node and a node as a link. In one or more embodiments according to the present invention, the map information is used when the travel plan generation unit **103** generates a travel plan.

[0030] The charging facility database **120** stores charging facility information related to one or more charging facilities. The charging facility information includes at least identification information of the charging facility, position information indicated by longitude and latitude, and charging availability information. The charging facility database **120** stores the charging facility information for each charging facility. The charging availability information includes full/vacant information indicating a vacant state of the charging station in the charging facility. Specifically, the charging availability information includes the full/vacant information indicating the vacant state at a predetermined timing for each charging station disposed in the charging facility. In addition, the full/vacant information includes information related to the reservation for use of the charging station. Specifically, the full/vacant information includes information on a time in which a reservation for use is made for each charging station of the charging facility. In addition, the full/vacant information includes information on the charging time of the vehicle in a case in which there is a vehicle being charged for each charging station of the charging facility. The charging availability information is acquired via the communication device **130** for each charging facility and is updated in real time.

[0031] The communication device **130** transmits and receives information between the onboard communication device **250** of the vehicle **20** via the network. The communication device **130** transmits the travel plan information to the vehicle **20**. The communication device **130** also receives the vehicle information from the vehicle **20**. In addition, the communication device **130** receives the charging availability information from the charging facility.

[0032] Next, the vehicle **20** will be described. The vehicle **20** includes at least an onboard controller **200**, a subject vehicle position detection device **210**, a charging state acquisition device **220**, a drive device **230**, an output device **240**, and an onboard communication device **250**. In one or more embodiments according to the present invention, the vehicle **20** is assumed to be a vehicle capable of autonomous traveling, but the present invention is not limited, and may be a vehicle operated by a driver.

[0033] The onboard controller **200** is a computer configured to control each unit of the vehicle **20** and includes a ROM (Read Only Memory storing a program), a CPU (Central Processing Unit) executing the program stored in the ROM, and a RAM (Random Access Memory) that functions as an accessible storage device. For example, the onboard controller **200** acquires various kinds of information from the subject vehicle position detection device **210** and the charging state acquisition device **220**, and transmits the various kinds of information to the travel plan generation device **10** via the onboard communication device **250**. In addition, the onboard controller **200** acquires the travel plan information from the onboard communication device **250**, and outputs a signal of a target control amount for controlling the travel of the vehicle **20** to the drive device **230**.

[0034] The subject vehicle position detection device **210** detects the present position of the vehicle **20**, and can use, for example, a GPS device. The subject vehicle position detection device **210** acquires position information of the vehicle **20** by receiving radio waves transmitted from a plurality of satellite communications by a receiver. Further, the subject vehicle position detection device **210** can detect a change in the position information of the vehicle **20** by periodically receiving radio waves transmitted from a plurality of satellite communications. The acquired position information of the vehicle **20** is transmitted to the travel plan generation device **10**. In one or more embodiments according to the present invention, the position information on the present position of the vehicle **20** is used when the vehicle **20** travels along the travel route.

[0035] The charging state acquisition device **220** acquires battery information indicating the remaining battery capacity of the battery mounted on the vehicle **20**. The acquired battery information is transmitted to the travel plan generation device **10** via the onboard communication device **250**.

[0036] The drive device **230** executes travel control including acceleration/deceleration and steering of the vehicle **20** in order to travel in accordance with the travel plan acquired from the travel plan generation device **10**. The drive device **230** includes, for example, an electric motor and/or an internal combustion engine, which are travel drive sources, a power transmission device including a drive shaft and an automatic transmission that transmit outputs from the travel drive sources to drive wheels, and a drive mechanism such as a braking device that brakes the wheels. In addition, the driving device **230** may include other devices necessary for the vehicle **20** to travel, such as a headlight, a direction indicator, a hazard lamp, and a wiper.

[0037] The output device **240** outputs various kinds of information to the occupant of the vehicle **20**. The output device **240** is, for example, a speaker that outputs various types of information. The output device **240** may be configured by a display. In one or more embodiments according to the present invention, when there is no charging facility at which the charging of the vehicle **20** is available within the allowable waiting time period from the estimated arrival time, the output device **240** outputs the notification information indicating that there is no charging facility at which the charging of the vehicle **20** is available within the allowable waiting time period from the estimated arrival time.

[0038] The onboard communication device **250** transmits and receives information to and from the communication device **130** of the travel plan generation device **10** via the network. The onboard communication device **250** receives the travel plan information of the vehicle **20** from the communication device **130**. In addition, the onboard communication device **250** transmits the vehicle information, that is, the position information and the battery information of the vehicle **20**, to the communication device **130**.

[0039] Next, an example of a flowchart of a travel plan generation method executed by the travel plan generation device **10** having the above-described configuration will be described with reference to FIG. **3**. In one or more embodiments according to the present invention, when the occupant of the vehicle **20** inputs the destination information of the vehicle **20**, the control flow is executed from step S**1**.

[0040] In step S**1**, the controller **100** generates a travel plan in which the vehicle **20** travels from the present position to the destination via the first charging facility. Specifically, the controller **100** generates the travel plan based on the position information of the vehicle **20**, the destination information, and the position information of the first charging facility.

[0041] In step S**2**, the controller **100** starts the travel of the vehicle **20** based on the generated travel plan. Specifically, the controller **100** transmits the travel plan information to the vehicle **20** and controls the drive device **230** of the vehicle **20** to drive the vehicle **20** based on the transmitted travel plan.

[0042] In step S**3**, the controller **100** determines whether or not a predetermined timing has come until the vehicle **20** arrives at the first charging facility after the vehicle **20** starts traveling toward the first charging facility. For example, the controller **100** determines, as the predetermined timing, whether or not the position of the vehicle **20** has reached the position of the predetermined distance from the first charging facility based on the position information of the vehicle **20** and the map information. When determining that the predetermined timing has come, the process proceeds to step S**4**. When determining that the predetermined timing has not come, the process returns to step S**3**, and the process is repeated until the predetermined timing has come.

[0043] In step S**4**, the controller **100** acquires charging availability information of the first charging facility. For example, the controller **100** acquires full/vacant information of the first charging facility at the predetermined timing from the charging facility database **120**. At this time, the controller **100** may acquire the full/vacant information of the first charging facility at the scheduled arrival time at which the vehicle **20** arrives at the first charging facility.

[0044] In step S**5**, the controller **100** determines whether or not the charging of the vehicle **20** is available at the first charging facility based on the charging availability information of the first charging facility acquired in step S**4**. The controller **100** determines that the charging of the vehicle **20** is not available at the first charging facility when acquiring, as the charging availability information, the full/vacant information indicating that there are no charging stations available in the first charging facility. When determining that the charging of the vehicle **20** is not available at the first charging facility, the process proceeds to step S**6**. When determining that the charging of the vehicle **20** is available at the first charging facility, the flow ends.

[0045] In step S**6**, the controller **100** searches for a second charging facility as a replacement facility for the first charging facility. For example, the controller **100** searches for the second charging facility that is within a predetermined distance from the first charging facility, and that can charge the vehicle **20** at the estimated arrival time at which the vehicle **20** arrives.

[0046] In step S**7**, the controller **100** determines, based on the search result of the second charging facility, whether or not the charging of the vehicle **20** is available at the second charging facility at the estimated arrival time at which the vehicle **20** arrives at the second charging facility. When the charging of the vehicle **20** is available at the second charging facility at the estimated arrival time, the process proceeds to step S**8**. When the charging of the vehicle **20** is not available at the second charging facility at the estimated arrival time, the process proceeds to step S**10**.

[0047] In step S**8**, the controller **100** generates a travel plan in which the vehicle **20** travels from the present position to the destination through the second charging facility. Specifically, the controller **100** generates the travel plan based on the position information of the vehicle **20**, the destination information, and the position information of the second charging facility.

[0048] In step S**9**, the controller **100** starts the control of the travel of the vehicle **20** in accordance with the travel plan after generating the travel plan in which the vehicle **20** passes through the second charging facility. Specifically, the controller **100** transmits the travel plan information to the vehicle **20**, and controls the drive device **230** of the vehicle **20** based on the travel plan for traveling of the vehicle **20**.

[0049] In one or more embodiments according to the present invention, after changing the charging facility through which the vehicle **20** passes from the first charging facility to the second charging facility, the controller **100** may acquire the charging availability information of the second charging facility at a predetermined timing while the vehicle **20** is traveling toward the second charging facility, and determine whether or not the charging of the vehicle **20** is available at the second charging facility. In this case, when determining that the charging of the vehicle **20** is not available at the second charging facility, the controller **100** searches for a charging facility different from the second charging facility again and generates a new travel plan.

[0050] In step S**10**, the controller **100** determines whether or not there is a charging facility at which the charging of the vehicle **20** becomes available within the allowable waiting time period from the estimated arrival time at which the vehicle **20** arrives at the charging facility. Specifically, the controller **100** determines whether or not the charging of the vehicle **20** becomes available at the first charging facility or the second charging facility within the allowable waiting time period from the estimated arrival time. Further, the controller **100** may determine whether or not the charging of the vehicle **20** becomes available within the allowable waiting time period from the estimated arrival time for each charging facility located in a predetermined range from the first charging facility. When there is a charging facility at which the charging of the vehicles **20** becomes available within the allowable waiting time period from the estimated arrival time, the process proceeds to step S**11**. When there is no charging facility at which the charging of the vehicles **20** becomes available within the allowable waiting time period from the estimated arrival time, the process proceeds to step S**12**. Further, in step S**10**, the controller **100** may compare the time periods until the charging of the vehicle **20** becomes available at each charging facility, and select a charging facility at which the charging of the vehicle **20** becomes available more quickly.

[0051] In step S**11**, the controller **100** generates a travel plan in which the vehicle **20** arrives at the first charging facility or the second charging facility after a lapse of time period from the estimated arrival time until the charging of the vehicle **20** becomes available. For example, the controller **100** controls the vehicle **20** to travel to a predetermined standby place, stop at the standby place for a period of time until the charging of the vehicle **20** becomes available. Then the controller **100** controls the vehicle to travel to the first charging facility or the second charging facility. This will allow the vehicle **20** to arrive at the first charging facility or the second charging facility after a lapse of time period from the estimated arrival time until the charging of the vehicle **20** becomes available.

[0052] In step S**12**, the controller **100** controls the output device **240** of the vehicle **20** to execute notification control. Specifically, the controller **100** controls the output device **240** to notify notification information indicating that there is no charging facility at which the charging of the vehicle **20** becomes available within the allowable waiting time period from the estimated arrival time.

[0053] As described above, the travel plan generation device and travel plan generation method according to the present embodiment generate a travel plan in which a vehicle passes through a first supply facility for supplying energy of the vehicle, acquire supply availability information on the energy of the first supply facility at a predetermined timing after the vehicle starts traveling based on the generated travel plan, determine whether or not supply of the energy is available at the first supply facility based on the supply availability information, when determining that the supply of the energy is not available at the first supply facility, identify a second supply facility different from the first supply facility and generate the travel plan in which the vehicle passes through the second supply facility instead of the first supply facility. Accordingly, the charging facility through which the vehicle passes can be changed when the supply of the energy for the vehicle becomes not available at the charging facility after the vehicle starts traveling until the vehicle arrives at the charging facility through which the vehicle is scheduled to pass.

[0054] Furthermore, the travel plan generation device and travel plan generation method according to the present embodiment set, as the predetermined timing, a time point at which the vehicle has reached a position within a predetermined range from the first supply facility. This enables the controller to grasp whether or not the supply of the energy at the first supply facility is available when the vehicle approaches the first supply facility.

[0055] Furthermore, the travel plan generation device and travel plan generation method according to the present embodiment set an interval between the predetermined timing to a predetermined interval. This enables the controller to grasp whether or not the supply of the energy at the first supply facility is available at a predetermined interval.

[0056] Furthermore, the travel plan generation device and travel plan generation method according to the present embodiment acquire, as the first supply availability information, full/vacant information indicating a vacancy state of a charging station in the first supply facility at the predetermined timing. This enables the controller to determine whether or not the supply of the energy at the first charging facility is available based on the vacancy state of the charging station in the first charging facility at the predetermined timing.

[0057] Furthermore, the travel plan generation device and travel plan generation method according to the present embodiment acquire, as the first supply availability information, full/vacant information indicating a vacancy state of a charging station at the first supply facility at a scheduled arrival time at which the vehicle arrives at the first supply facility. This enables the controller to determine whether or not the supply of the energy at the first charging facility is available based on the vacancy state of the charging station in the first charging facility at the time when the vehicle arrives at the first charging facility.

[0058] Furthermore, the travel plan generation device and travel plan generation method according to the present embodiment acquire second supply availability information of the second supply facility after an estimated arrival time at which the vehicle arrives at the second supply facility, determine whether or not the supply of the energy is available at the second supply facility based on the second supply availability information, and when determining that the supply of the energy is available at the second supply facility, generate the travel plan in which the vehicle arrives at the second supply facility after a lapse of time period from the estimated arrival time until the supply of the energy becomes available. This enables the controller to adjust the arrival time of the vehicle at a timing at which the supply of the energy becomes available at the second supply facility.

[0059] Furthermore, the travel plan generation device and travel plan generation method according to the present embodiment acquire first supply availability information of the first supply facility after an estimated arrival time at which the vehicle arrives at the first supply facility, determine whether or not the supply of the energy is available at the first supply facility based on the first supply availability information and when determining that the supply of the energy is available at the first supply facility, generate the travel plan in which the vehicle arrives at the first supply facility after a lapse of time period from the estimated arrival time until the supply of the energy becomes available. This enables the controller to adjust the arrival time of the vehicle at a timing at which the supply of the energy becomes available at the first supply facility.

[0060] Furthermore, the travel plan generation device and travel plan generation method according to the present embodiment select a standby place for the vehicle to standby from a standby available place registered in advance in map information and generate the travel plan in which the vehicle stops at the standby place for a time period until the supply of the energy becomes available. This enables the controller to adjust the time of arrival at the first supply facility or the second supply facility by stopping the vehicle at the standby place.

[0061] It should be appreciated that the embodiments explained heretofore are described to facilitate understanding of the present invention and are not described to limit the present invention. It is therefore intended that the elements disclosed in the above embodiments include all design changes and equivalents to fall within the technical scope of the present invention.

DESCRIPTION OF REFERENCE NUMERALS

[0062] **10** . . . . Travel plan generation device
[0063] **100** . . . . Controller
[0064] **101** . . . . Vehicle information acquisition unit
[0065] **102** . . . . Charging availability information acquisition unit
[0066] **103** . . . . Travel plan generation unit
[0067] **104** . . . . Charging availability determination unit
[0068] **105** . . . . Notification control unit
[0069] **110** . . . . Map database
[0070] **120** . . . . Charging facility database
[0071] **130** . . . . Communication device
[0072] **20** . . . . Vehicle
[0073] **200** . . . . Onboard controller
[0074] **210** . . . . Subject vehicle position detection device
[0075] **220** . . . . Charging state acquisition device
[0076] **230** . . . . Drive device
[0077] **240** . . . . Output device
[0078] **250** . . . . Onboard communication device

What is claimed is:

1. A travel plan generation device comprising:
a controller, the controller is configured to:
generate a travel plan in which a vehicle passes through a first supply facility for supplying energy of the vehicle;
acquire first supply availability information indicating a vacancy state of a first supply facility after the vehicle starts traveling based on the generated travel plan;
notify notification information indicating that there is no available charging station;
determine whether or not supply of the energy is available at the first supply facility based on the first supply availability information of the first supply facility; and
when determining that the supply of the energy is not available at the first supply facility, identify a second supply facility different from the first supply facility and generate the travel plan in which the vehicle passes through the second supply facility instead of the first supply facility.

2. The travel plan generation device according to claim **1**, wherein the controller is further configured to notify that the supply of the energy is not available at the first supply facility when determining that the supply of the energy is not available at the first supply facility.

3. The travel plan generation device according to claim **1**, wherein the controller is further configured to acquire the first supply availability information when the vehicle arrives at a position within a predetermined range from the first supply facility at a predetermined timing after the vehicle starts traveling based on the generated travel plan.

4. The travel plan generation device according to claim **1**, wherein the controller is further configured to:
acquire second supply availability information of the second supply facility after an estimated arrival time at which the vehicle arrives at the second supply facility;
determine whether or not the supply of the energy is available at the second supply facility based on the second supply availability information; and
when determining that the supply of the energy is available at the second supply facility, generate the travel plan in which the vehicle arrives at the second supply facility after a lapse of time from the estimated arrival time until the supply of the energy becomes available.

5. The travel plan generation device according to claim **1**, wherein the controller is further configured to:

acquire the first supply availability information of the first supply facility after an estimated arrival time at which the vehicle arrives at the first supply facility;

determine whether or not the supply of the energy is available at the first supply facility based on the first supply availability information; and

when determining that the supply of the energy is available at the first supply facility, generate the travel plan in which the vehicle arrives at the first supply facility after a lapse of time from the estimated arrival time until the supply of the energy becomes available.

6. The travel plan generation device according to claim **4**, wherein the controller is further configured to select a standby place for the vehicle to standby from a standby available place registered in advance in map information and generates the travel plan in which the vehicle stops at the standby place for a time period until the supply of the energy becomes available.

7. A travel plan generation device comprising:

a controller, the controller is configured to:

generate a travel plan in which a vehicle passes through a first supply facility for supplying energy of the vehicle;

acquire first supply availability information on the energy of a first supply facility after the vehicle starts traveling based on the generated travel plan;

notify notification information indicating that there is no available charging station;

determine whether or not supply of the energy is available at the first supply facility based on the first supply availability information of the first supply facility; and

when determining that the supply of the energy is not available at the first supply facility, identify a second supply facility different from the first supply facility and generate the travel plan in which the vehicle arrives at one of the first and second supply facilities in which the supply of the energy becomes available within an allowable waiting time from an estimated arrival time after a lapse of time from the estimated arrival time until the supply of the energy becomes available.

8. A travel plan generation method executed by a controller, the method comprises:

generating a travel plan in which a vehicle passes through a first supply facility for supplying energy of the vehicle;

acquiring supply availability information indicating a vacancy state of the first supply facility after the vehicle starts traveling based on the generated travel plan;

notifying notification information indicating that there is no available charging station;

determining whether or not supply of the energy is available at the first supply facility based on the supply availability information;

when determining that the supply of the energy is not available at the first supply facility, identifying a second supply facility different from the first supply facility; and

generating the travel plan in which the vehicle passes through the second supply facility instead of the first supply facility.

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